



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Congruence & Similarity

Contents

- * Congruence
- * Similarity
- * Similar Lengths
- * Similar Areas & Volumes



Your notes

Congruence

Congruence

What is congruence?

- Two shapes are **congruent** if they are **identical** in **shape** and **size**
 - One may be a **reflection**, **rotation**, or **translation** of the other
- If one shape is an **enlargement** of the other, then they are **not identical in size** and so are **not** congruent

How do we prove that two shapes are congruent?

- To show that two shapes are **congruent** you need to show that they are both the **same shape** and the **same size**
 - If a shape has been reflected, rotated or translated, then its image is **congruent** to it
- Show that **corresponding sides** are the **same length**
- Show that **corresponding angles** are the **same size**
- You do **not** need to show that they are facing in the **same direction**



Examiner Tips and Tricks

- Tracing paper** can help in the exam if you are unsure whether two shapes are congruent
 - Trace over one shape and then see if it fits **exactly** on top of the other
 - Only do this if the image is drawn to **scale**



Worked Example

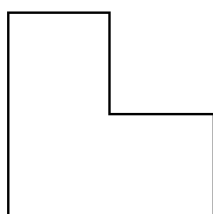
Write down the letters of the two shapes below which are congruent to A.



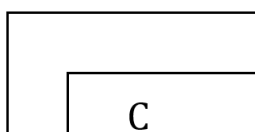
Your notes



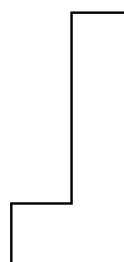
A



B



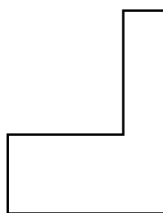
C



D



E



F

Shapes C and D are congruent to A



Your notes

Similarity

Similarity

What are similar shapes?

- Two shapes are **similar** if they have the same **shape** and their **corresponding sides** are in **proportion**
 - One shape is an **enlargement** of the other

How do we prove that two triangles are similar?

- To show that two **triangles** are **similar** you need to show that their **angles are the same**
 - If the angles are the same then **corresponding lengths** of a triangle will automatically be in **proportion**
- You can use angle properties to **identify equal angles**
 - Look out for **isosceles triangles**, **vertically opposite angles** and **angles on parallel lines**
- If a question asks you to **prove two triangles** are **similar**
 - For **each pair** of corresponding angles
 - State** that they are of **equal size**
 - Give a **reason** for why they are equal

How do we prove that two shapes are similar?

- To show that two **non-triangular shapes** are **similar** you need to show that their **corresponding sides** are in **proportion**
 - Divide the length of one side by the length of the corresponding side on the other shape to find the **scale factor**
- If the **scale factor** is the **same** for **all corresponding sides**, then the shapes are **similar**



Examiner Tips and Tricks

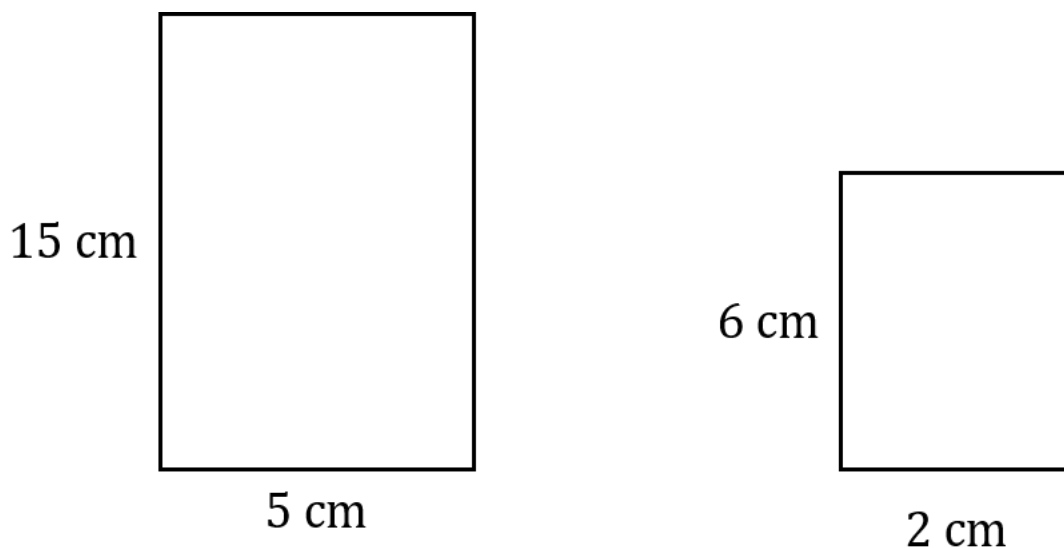
- A pair of **similar triangles** can often be **opposite** each other in an hourglass formation.
 - Look out for the **vertically opposite**, equal angles.

- It may be helpful to sketch the triangles **next** to each other and facing in the **same direction**.



Worked Example

(a) Prove that the two rectangles shown in the diagram below are similar.



Use the corresponding lengths (15 cm and 6 cm) to find the scale factor

$$\frac{15}{6} = 2.5$$

Use the corresponding width (5 cm and 2 cm) to find the scale factor for the other pair of sides

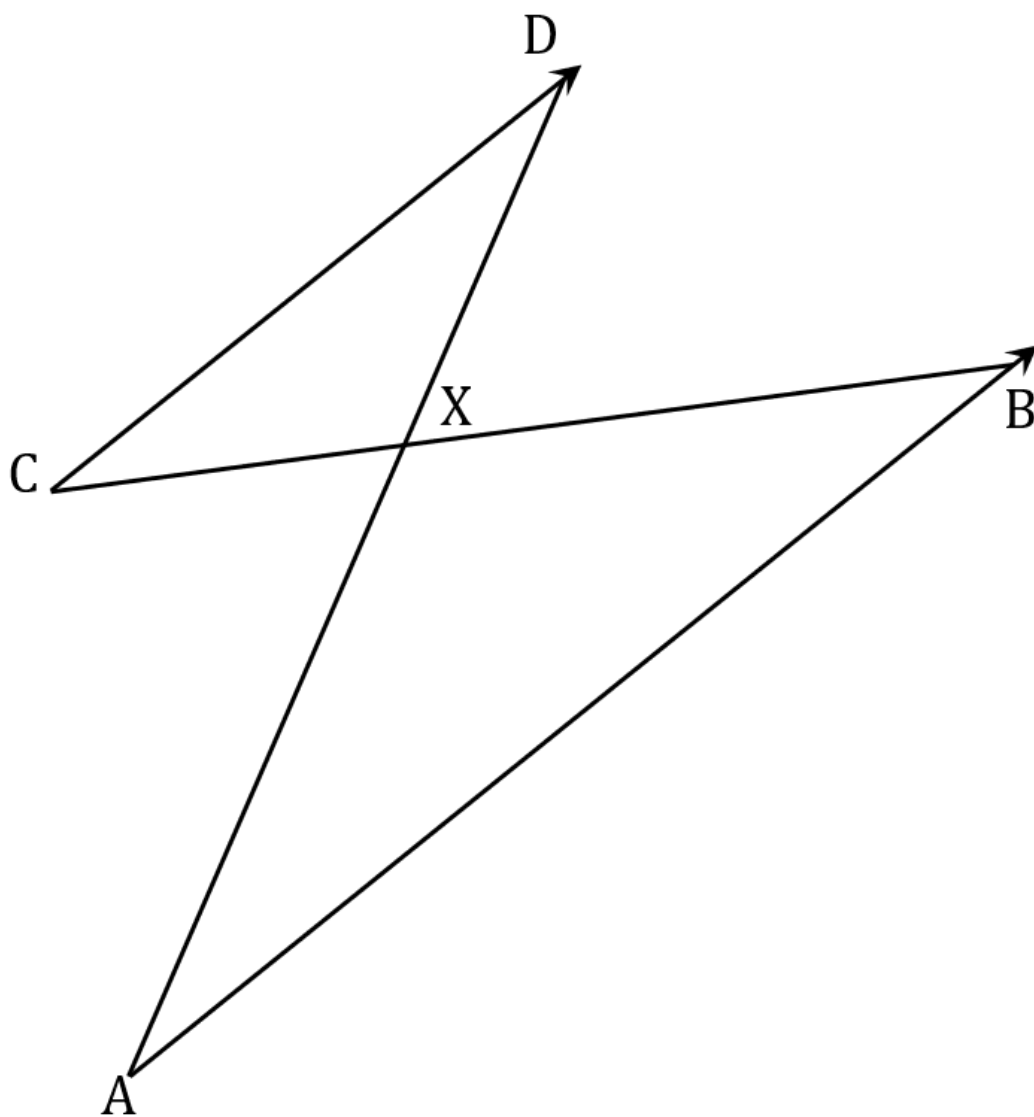
$$\frac{5}{2} = 2.5$$

The two rectangles are similar, with a scale factor of 2.5

(b) In the diagram below, AB and CD are parallel lines.
 Show that triangles ABX and CDX are similar.



Your notes



State the equal angles by name, along with clear reasons

Don't forget to state that similar triangles need to have equal corresponding angles

Angle AXB = angle CXD (vertically opposite angles are equal)

Angle ABC = angle BCD (alternate angles on parallel lines are equal)

Angle BAD = angle ADC (alternate angles on parallel lines are equal)

All three corresponding angles are equal, so the two triangles are similar



Your notes

Similar Lengths

Similar Lengths

How do I solve problems that involve similar lengths?

- Equivalent **lengths** in two similar shapes will be in the same ratio and are linked by a **scale factor**
 - Identify **known** lengths of **corresponding sides**
 - Establish the **type** of enlargement
 - If the shape is **getting bigger**, then the **scale factor** is **greater than 1**
 - If the shape is **getting smaller**, then the **scale factor** is **greater than 0** but **less than 1**
 - Find the **scale factor**
 - **Divide** a **known length** on the **second shape** by the **corresponding known length** on the **first shape**
 - Use the scale factor to find the **length** you need
 - **Multiply** a known length by the **scale factor** on the **first shape** to find the **corresponding length** on the **second shape**
 - **Divide** a known length on the **second shape** by the **scale factor** to find the **corresponding length** on the **first shape**

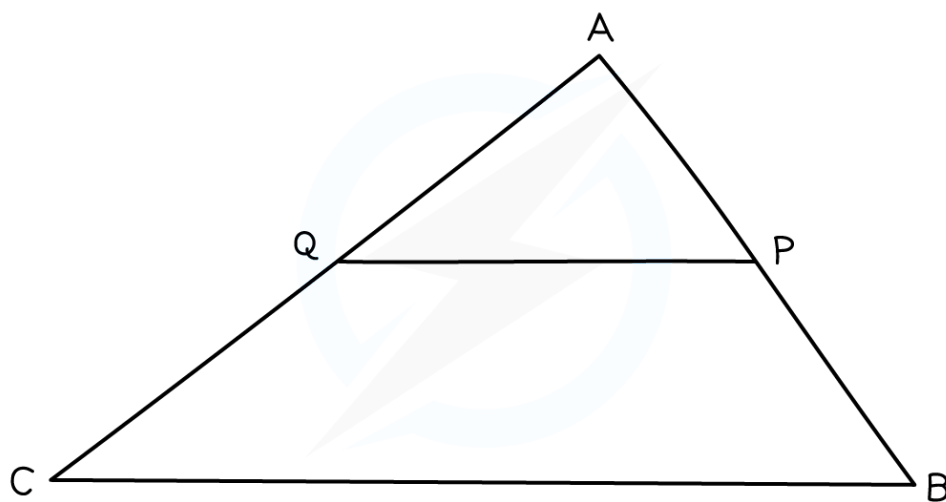


Examiner Tips and Tricks

- If similar shapes overlap on the diagram (or are not clear) draw them separately.
 - For example, in this diagram the triangles ABC and APQ are similar:



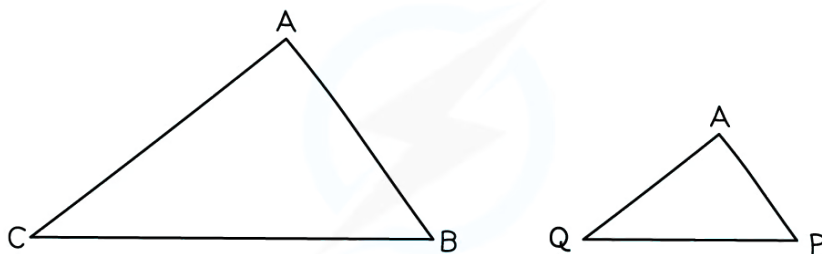
Your notes



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- So redraw them separately before starting:



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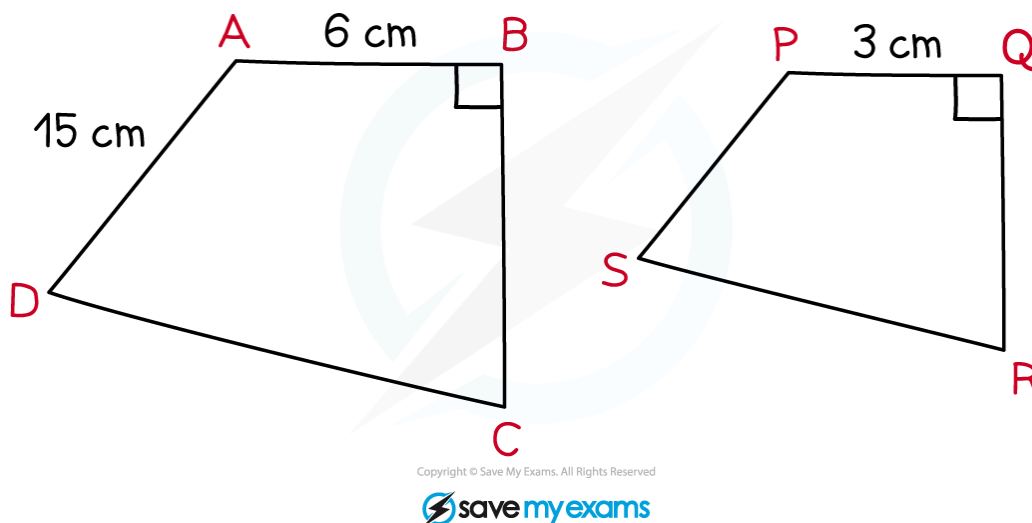


Worked Example

$ABCD$ and $PQRS$ are similar shapes.
Find the length of PS .



Your notes



The two shapes are mathematically similar

Each length on the first shape can be multiplied by a scale factor to find the corresponding length on the second shape

Identify two known corresponding sides of the similar shapes

AB and PQ are corresponding sides

The second shape is smaller than the first shape so the scale factor will be between 0 and 1

Divide the known length on the second shape by the corresponding length on the first shape to find the scale factor

$$\text{Scale Factor} = \frac{3}{6} = \frac{1}{2}$$

Multiply the length AD by the scale factor to find its corresponding length PS on the second shape

$$PS = \frac{1}{2} \times 15 = \frac{15}{2}$$

$$PS = 7.5 \text{ cm}$$



Your notes

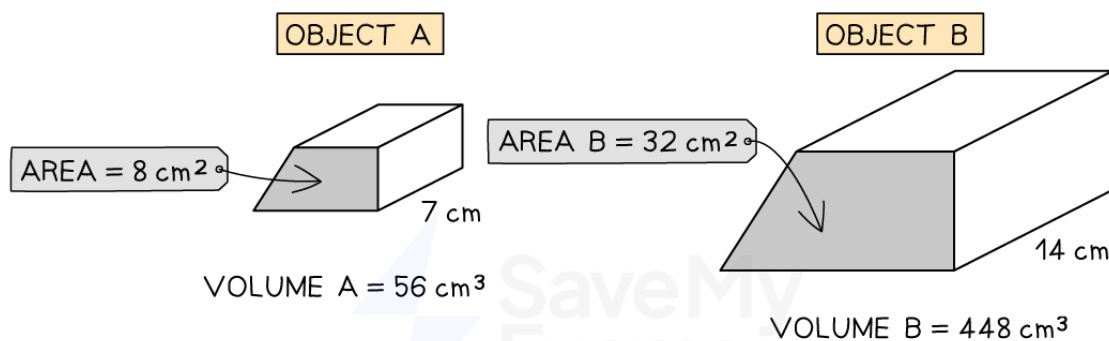
Similar Areas & Volumes

Similar Areas & Volumes

What are similar shapes?

- Two shapes are **mathematically similar** if one is an **enlargement** of the other
- If the **lengths** of two similar shapes are linked by the **scale factor**, k
 - Equivalent **areas** are linked by an **area factor**, k^2
 - Equivalent **volumes** are linked by a **volume factor**, k^3
- The **scale factor** (SF) for a given quantity (length, area or volume), can be found using the formula:

$$\text{scale factor} = \frac{\text{second quantity}}{\text{first quantity}}$$



$$\text{LENGTH SF} = \frac{14}{7} = 2$$

$$\text{AREA SF} = \frac{32}{8} = 4$$

$$\text{VOLUME SF} = \frac{448}{56} = 8$$

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- An object could be made either **bigger or smaller** by a scale factor
 - When $k > 1$, the object is getting bigger
 - This is also true for $k^2 > 1$ and $k^3 > 1$
 - When $0 < k < 1$, the object is getting smaller

- This is also true for $0 < k^2 < 1$ and $0 < k^3 < 1$

How do I work with similar shapes involving area or volume?



Your notes

- **STEP 1**

Identify the **equivalent** known quantities

- Recognise if the quantities are lengths, areas or volumes

- **STEP 2**

Find the **scale factor** from two known **lengths**, **areas** or **volumes**

- $$\text{scale factor} = \frac{\text{second quantity}}{\text{first quantity}}$$

- For two lengths, $k = \text{length SF}$
- For two areas, $k^2 = \text{area SF}$
- For two volumes, $k^3 = \text{volume SF}$

- **STEP 3**

Check the scale factor

- $\text{SF} > 1$ if getting bigger
- $0 < \text{SF} < 1$ if getting smaller

- **STEP 4**

If necessary, use the scale factor you have found to find **other scale factors**

- If you have the **length** scale factor
 - $\text{Area scale factor} = (\text{Length scale factor})^2$
 - $\text{Volume scale factor} = (\text{Length scale factor})^3$
- If you have the **area** scale factor
 - $\text{Length scale factor} = \sqrt{(\text{Area scale factor})}$
 - Find the volume scale factor by finding the length scale factor first
- If you have the **volume** scale factor
 - $\text{Length scale factor} = \sqrt[3]{(\text{Volume scale factor})}$
 - Find the area scale factor by finding the length scale factor first



Your notes

STEP 5Multiply or divide by relevant scale factor to find a **new quantity****Examiner Tips and Tricks**

- Take extra care not to mix up **which shape is which** when you have started carrying out the calculations
 - It can help to **label the shapes** and write an equation

**Worked Example**

Solid A and solid B are mathematically similar.

The volume of solid A is 32 cm^3 .The volume of solid B is 108 cm^3 .

The height of solid A is 10 cm.

Find the height of solid B.

Calculate k^3 , the scale factor of enlargement for the volumes, using:

$$\text{volume } B = k^3(\text{volume } A)$$

$$\text{Or } k^3 = \frac{\text{larger volume}}{\text{smaller volume}}$$

$$108 = 32k^3$$

$$k^3 = \frac{108}{32} = \frac{27}{8}$$

Find the length scale factor k by taking the cube root of the volume scale factor k^3

$$k = \sqrt[3]{\frac{27}{8}} = \frac{3}{2}$$

Substitute the value for k into formula for the heights of the similar shapes:

$$\text{Height } B = k(\text{height } A)$$

$$h = 10k$$

$$h = 10\left(\frac{3}{2}\right) = \frac{30}{2} = 15$$

Height of $B = 15$ cm



Your notes